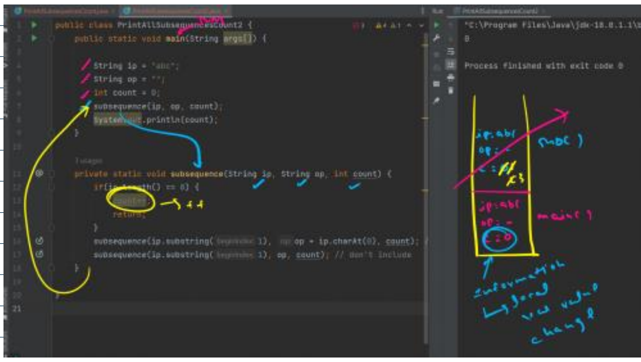
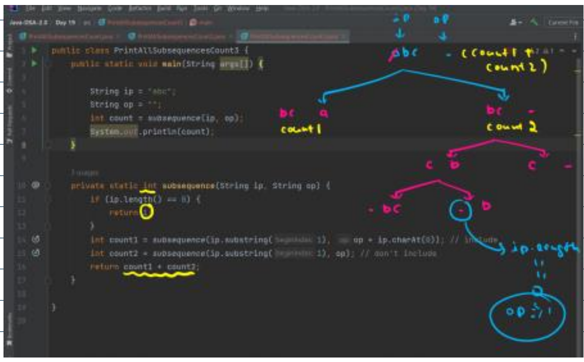
Recursion

"Recursion works like magic"

f(n) → $\boxed{f(n-1)}$

↓
Trust that
it will
return a
valid op





Print all possible outcome of a flip

1 coin :- H / T
2 coin :- H H
H T
T H
T T

3 coin :- H H H
H H T
H T H
H T T
T H H
T H T
T T H
T T T

all possible outcomes

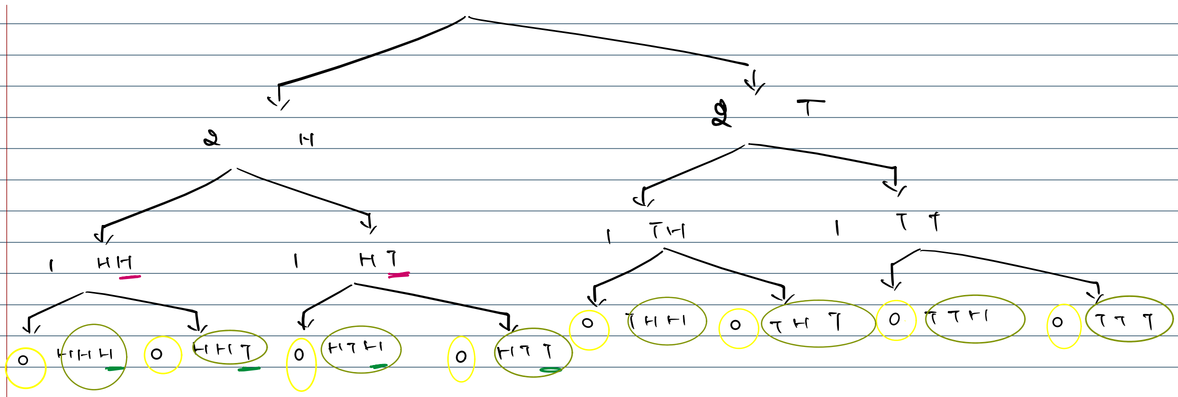
Input :- no. of coins

output

Recursion :- choices + Decision

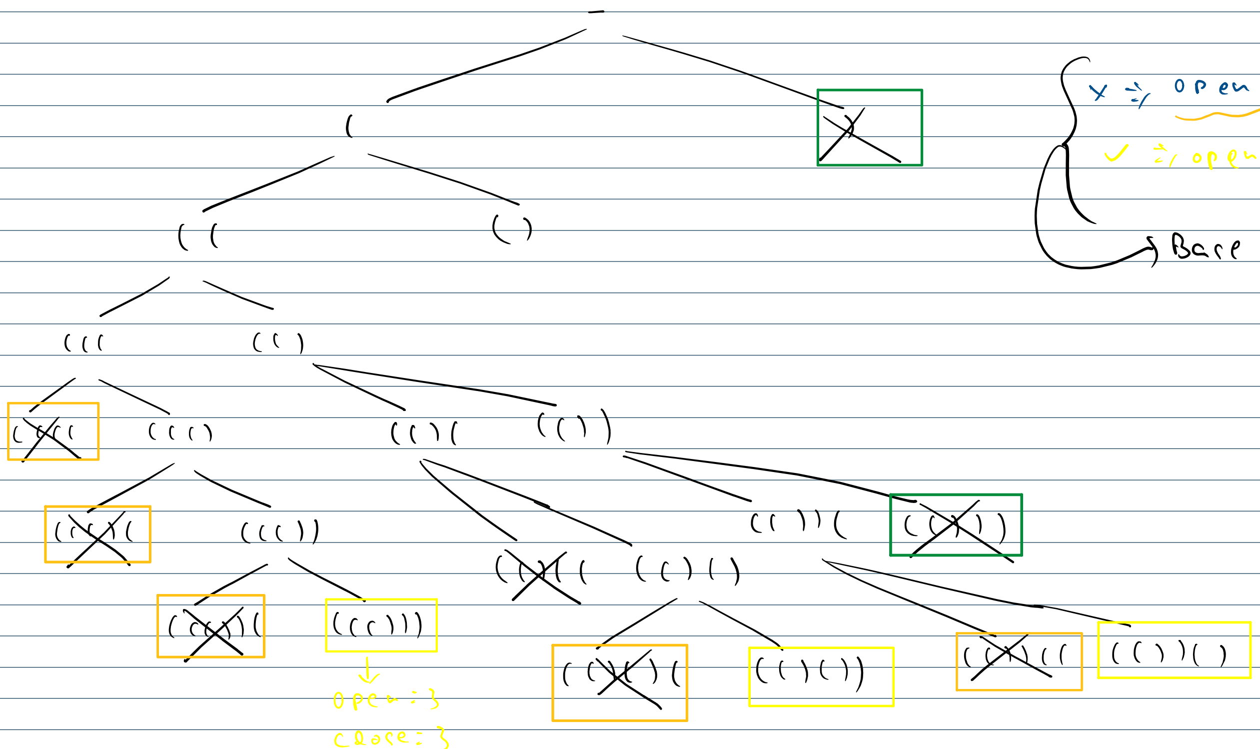
→ H
→ T

ip op
↓ ↓
3 _



string: 0 => ans => Base condition => smallest valid ip.

Generate all paranthesis



X => open > n q closed > ope
✓ => open == n && closed =
Base condition.

Print all permutation of a String

permutation ⟶ arrangements

(A) (B) (C) =>  => students

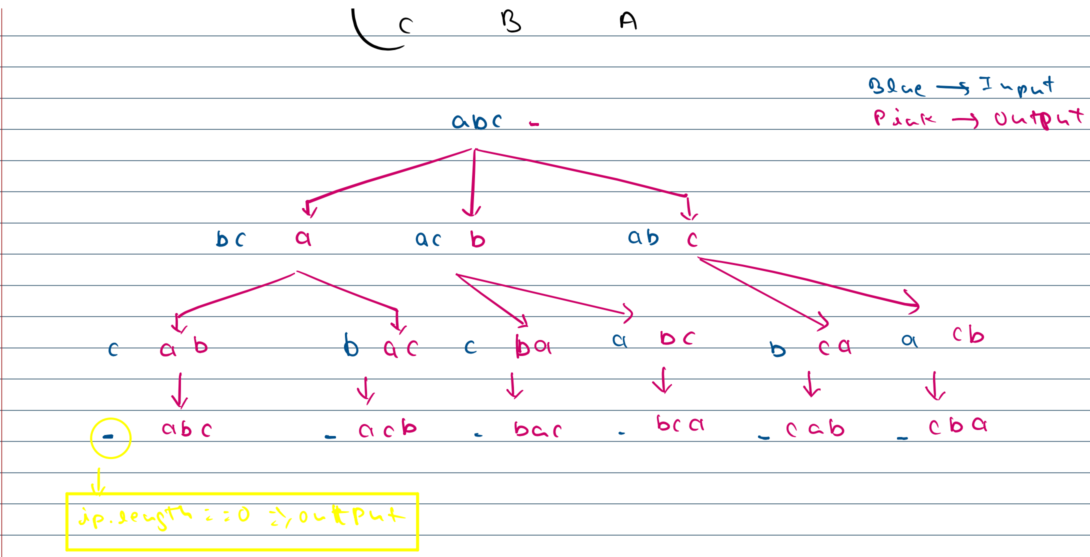
6 perm

A	B	C
A	C	B
B	A	C
B	C	A
C	A	B
C	B	A

Blue → Input

2 n

: n



0 1 2 3 4 5
a b c d e f → ans = ans + s.charAt(3)
↓
abcef
↓
Input
abc + ef
↓
sub(0, 3) + sub(4, 6) ⇒ Input.

0 1 2 3 4 5
a b c d e f → ans = ans + s.charAt(4)
↓
abcdf
↓
abcd + f
↓
sub(0, 4) + sub(5, 6)
↓
Input

Repeat the same job with length of string